Plant Location

Theory Notes & Summary of Location Models

A condensed summary of the theoretical concepts from the presentation 'Plant Location Models' — covering the need for location decisions, the factors influencing them, the major quantitative location models, and the design considerations (equipment, space, labour) that follow a location decision.

1. Introduction to Plant Location

Plant (facility) location is an important **strategic-level** decision for an organisation. A key feature of a conversion/manufacturing process is the efficiency with which products or services reach customers — and this depends heavily on where the plant is placed. Before a location is finalised, long-range forecasts of the company's future needs should be made.

2. Need for Selecting a Suitable Location

The need for a location decision arises in three situations:

- **New organisation** — choosing a location for the first time.
- **Existing organisation** — expansion or relocation of an operating plant.
- **Global location** — setting up or extending operations internationally.

2.1 Location Choice for a New Organisation

Selecting a location for a new organisation involves three stages:

- **Identification of region** — Guided by organisational objectives and long-term considerations: marketing, technology, internal strengths/weaknesses, region-specific resources, business environment, legal-governmental environment, social environment, and geography.
- **Choice of a site within a region** — Once a region is fixed, the best site is chosen from the available set by evaluating the tangible and intangible costs of alternatives. This step depends less on long-term strategy.
- **Dimensional analysis** — Used when both tangible and intangible costs must be considered together. Relative merit (cost ratio) of each cost item is computed for two alternative sites, each ratio is raised to an assigned weight, and the weighted ratios are multiplied to get one comprehensive figure. If the resulting ratio is greater than 1, the comparison site is superior, and vice-versa.

Because plant relocation later involves production stoppage, shifting costs and operational disruption, it is best to generate several alternative sites at the outset and select the best one after critical analysis.

2.2 Location Choice for an Existing Organisation

Here a manufacturing plant must fit into a firm's **multi-plant operations strategy**. Four broad strategies are used:

- **Plants manufacturing distinct products** — Each plant serves the entire market with its own product line; useful when products need distinctly different technology/resources or in a highly competitive market. Greater decentralisation improves planning, control and resource use.
- **Plants supplying a specific market area** — Each plant makes almost the full product range for its region; used when market proximity dominates over resource/technology considerations (e.g. soft-drink bottling). Requires heavy coordination from the corporate office.
- **Plants divided by process/manufacturing stage** — Each stage needs distinct equipment, labour skills and technology; output of one plant feeds the next, so centralised coordination is essential.

- **Plants emphasising flexibility** — Requires close coordination to meet changing needs while using facilities efficiently. The central question for any location decision is whether the company can remain competitive at that location over the long term.

2.3 Global Location

Globalisation has led multinational firms to set up in India, and Indian firms to expand abroad. Two concepts of global presence apply:

- **Virtual proximity** — Advances in telecommunications let a firm stay close to customers without physical presence; logistics and business transactions increasingly move through the information/communication pathway (e.g. software services firms).
- **Virtual factory** — A firm outsources part of its operations to a business associate's facility abroad instead of owning operations there (e.g. an Indian BPO acting as a foreign company's 'virtual service factory'). The location decision need not involve the firm's own operations.

Reasons for a Global/Foreign Location

Tangible reasons

- Reaching the customer directly.
- Tax advantages offered by the host country.
- Lower manufacturing/operating costs (labour, raw material, energy, water, key personnel).
- Overcoming tariff barriers by manufacturing locally instead of exporting.

Intangible reasons

- **Customer-related** — Greater accessibility and service quality; closer customer interaction and understanding; discovery of new potential customers.
- **Organisational learning-related** — Learning advanced technology, and learning from customers, competitors and suppliers through physical presence.
- **Other strategic reasons** — 'Local' psychological advantage; reduced political risk via multi-country presence; alternative supply sources; access to global human capital; lower overall market risk by spreading across markets.

3. Location Models

Several quantitative models help identify the ideal location. The five popular models are:

- Factor Rating Method
- Weighted Factor Rating Method
- Load–Distance Method
- Centre of Gravity Method
- Locational Break-Even Analysis

3.1 Factor Rating Method

Steps in selecting a new facility location:

- Identify the important location factors.
- Rate each factor by its relative importance (higher rating = more prominent factor).

- Score each candidate location against every factor.
- Calculate each location's rating by multiplying the assigned score by the factor rating.
- Sum the products for each location; the location with the **highest total score** is selected.

3.2 Weighted Factor Rating Method

An extension of factor rating: factors are assigned **weights** based on relative importance, and each site is scored against every factor (commonly on a 1 = poor to 5 = excellent scale). A weighted score is computed for each site using a preference matrix, and the site with the **highest weighted score** is selected as the best choice.

3.3 Load–Distance Method

A mathematical model that evaluates locations on **proximity**. The objective is to choose a location that minimises the total weighted load moving into and out of the facility. Locations are plotted as grid coordinates on a map, and distance (or, alternatively, time) between points is measured. A common distance measure is **rectilinear distance** — the distance between two points following a series of 90° turns (as along city blocks), i.e. the sum of the horizontal and vertical legs between the points. The load-distance score for a candidate site is the sum of (population/load × rectilinear distance) over all points served; the site with the **lowest total load-distance score** is preferred.

3.4 Centre of Gravity Method

Based primarily on **cost considerations**, this method helps balance cost and service objectives. It takes into account the locations of existing plants/markets, the volume of goods moved, and transportation costs to identify the best location for a single intermediate facility (e.g. a warehouse). The centre of gravity is computed as the load-weighted average of the x- and y-coordinates of all the demand points being served.

3.5 Locational Break-Even Analysis

Uses **cost–volume analysis** to economically compare location alternatives. By identifying the fixed and variable costs of each location and graphing the resulting total-cost lines, the location with the lowest total cost for a given expected production volume can be identified. The three steps are:

- Determine the fixed and variable cost for each location.
- Plot the total-cost line for each location (cost on the vertical axis, volume on the horizontal axis).
- Select the location with the **lowest total cost** for the expected production volume (different locations may be optimal over different volume ranges).

4. Factors Influencing Plant/Facility Location

Locational factors fall into two broad groups: **general locational factors** (controllable and uncontrollable, applicable to all organisation types) and **specific locational factors** (particular to manufacturing or service organisations).

4.1 Controllable Factors

- **Proximity to markets** — Preferred when products are delicate/perishable, after-sales service is frequently needed, transportation cost is high, or shelf life is short. Nearness to market ensures consistent supply and lower transport cost.
- **Supply of raw material** — Critical when materials are perishable or transport cost is high. General guidelines (Yaseen): locate near the raw-material source if the material loses weight in processing; locate near the market if the raw material is universally available; locate to minimise total transport cost if materials come from varied sources. Important for industries like sugar, cement, jute and cotton textiles.
- **Transportation facilities** — Speedy transport (air, road, rail, water, pipeline) ensures timely raw-material supply and finished-goods delivery. Export-oriented plants need proximity to a port/airport; transport cost relative to value added is a key location criterion.
- **Infrastructure availability** — Power, water and waste-disposal facilities are critical. Power-intensive industries (aluminium, steel) need assured uninterrupted power; process industries (paper, chemicals, cement) need large, good-quality water supply and proper waste disposal.
- **Labour and wages** — Availability of adequately skilled labour, cost of importing labour, local labour-relations history, labour productivity, prevailing wage levels, cost of living, and union bargaining power all matter.
- **External economies of scale** — Urbanisation economies (advantages of locating in a large city — access to labour, transport, markets, business services) and localisation economies (advantages of clustering near related industries).
- **Capital** — Fixed capital (building/construction) costs vary by region, though buildings can be rented or existing plants expanded. Financial capital is highly mobile and has little influence on location, since large multinationals can raise capital wherever conditions are most favourable.

4.2 Uncontrollable Factors

- **Government policy** — Labour laws, building codes and safety regulations demand attention; central/state governments may offer incentive packages (tax/excise exemptions, soft loans, electricity subsidies, investment subsidies) to promote balanced regional growth.
- **Climatic conditions** — Geology and climate (humidity, temperature) affect human efficiency and behaviour; some industries need specific conditions (e.g. textile mills need humidity).
- **Supporting industries and services** — Availability of vendors/subcontractors for components, plus communication, banking, consultancy and other civic services.
- **Community and labour attitudes** — Attitudes toward work, management and trade-union activity can make a location undesirable even if all other factors are favourable (risk of strikes/lockouts).
- **Community infrastructure and amenities** — Economic overhead capital (roads, railways, port facilities, power lines) and social overhead capital (schools, universities, hospitals).

5. Serviceability and Flexibility

5.1 Serviceability

Serviceability is the measure of how easily, quickly and frequently corrective and preventive maintenance can be carried out on a system — it directly affects the system's reliability and

availability. Good manageability improves serviceability.

5.2 Flexibility

Facility flexibility is mainly related to **multi-product processing** and **scalability**, with mobility and achievable product life-cycle as secondary factors.

Dimensions of flexibility:

- Material quality
- Output quality
- New product introduction
- Modification
- Deliverability
- Volume
- Product mix
- Resource mix

Advantages:

- Wide variety of products from the same facility
- More profitable investment
- Lower inventory requirement
- Lower labour cost
- Higher productivity
- Ability to meet emerging customer needs with little lead time
- Lower manufacturing cost and higher revenue

Disadvantages:

- Requires intensive planning
- Significant upfront implementation cost
- Can require major change in company culture

6. Selection of Equipment and Machines

Equipment selection depends mainly on the process followed by the industry, and choosing the right equipment is a prerequisite for low-cost production. Re-evaluation of equipment is triggered when production capacity must expand, product design changes radically, or technological upgrades are needed to stay competitive.

Main factors in selecting specific equipment:

- Kind of operations
- Required quality and precision
- Anticipated future changes in product design
- Volume of production (demand forecast)
- Availability of equipment relative to plant capacity
- Anticipated production rates of various-capacity machines

- Maintaining balanced capacity across equipment
- Type of layout (product or process layout)

Factors in selecting specific machines:

- First cost and suitability
- Constructability (schedule, lead time, commissioning)
- Ease and cost of operation and maintenance
- Total cost of ownership
- Manufacturer's experience and reputation
- Impact on other building-design elements
- Noise criteria and lifespan
- Energy efficiency / code requirements
- Scalability, staging and modularity
- Redundancy and failure-mode risk
- Environmental health attributes and safety

7. Space, Layout and Labour Planning

7.1 Space Requirements

Space needs depend on lot sizes, storage system, equipment type/size, layout arrangement, building configuration, housekeeping policies, material-handling equipment, and support spaces (offices, cafeteria, washrooms). Determining required space is one of the hardest parts of facilities planning; plans are typically projected 5–10 years ahead. Space is generally grouped into:

- **Equipment space** — the equipment itself, machine travel, machine maintenance, and plant services.
- **Materials space** — receiving/storing inbound materials, in-process materials, storing/shipping finished goods, storing/shipping waste and scrap, and tools/fixtures/dies.
- **Personnel space** — the operator, material handling, and operator ingress/egress.

7.2 Aisle Arrangement

Aisles should be positioned to promote effective flow through the facility, and aisle width should be set according to the type and volume of traffic it must carry.

7.3 Labour Requirements and Employee Facilities

Personnel planning covers employee parking, locker rooms, restrooms, food services, drinking fountains and health services.

Employee parking planning procedure:

- Determine the number of vehicles to be parked.
- Determine the space required per vehicle.
- Determine the space available for parking.
- Develop alternative parking layouts.

- Select the layout that best uses space and maximises employee convenience.

Food service options: dining away from the facility, vending machines with a cafeteria, a serving line with a cafeteria, or a full kitchen with a cafeteria.

Health facilities may include pre-employment examinations, first-aid treatment, major medical treatment, dental care and treatment of illnesses.

Summary compiled from the presentation 'Plant Location Models' (numerical worked examples and practice problems from the original slides are omitted; only the underlying theory is covered).